

Aleksei Zabelin

Forward Deployed Engineer / Solutions Engineer

Hoi An, Vietnam (UTC+7) · Open to relocation and client-site travel

Available across EU, UK, APAC, Australia and New Zealand hours

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SUMMARY

I take a customer problem from the first conversation to production, then stay responsible for it: discovery, a scoped plan, a proof of concept on real data, then the running system.

Sole engineer-owner of a virtual-tour SaaS on AWS serving luxury hospitality. I run the platform, build the LLM agents on top of it, and direct the developers who extend it. Twenty years delivering 360°/3D to enterprise clients — the last two replacing manual delivery work with retrieval-augmented generation and function-calling agents.

Portfolio — 360°/3D delivered for Ritz-Carlton · St Regis · Raffles · Atlantis The Palm · Shangri-La · Marriott · Six Senses · Anantara · Nihl Sumba · Joali Being

CORE STACK

Languages — Python, TypeScript, JavaScript, SQL, Bash

AI / LLM — Retrieval-augmented generation (RAG) over vector stores (Supabase / Postgres + pgvector), LLM agents, function calling / tool use, prompt engineering, AnythingLLM, self-hosted model deployment

Backend & Integration — Node.js, Express, REST API design, webhooks, data pipelines, third-party API integration, STS/OAuth flows

Cloud & DevOps — AWS (EC2, S3, CloudFront, IAM, STS, WAF, ACM, Route 53), Docker, GitHub Actions CI/CD, production monitoring and alerting, Cloudflare Workers, nginx, PM2, systemd

Data — MongoDB / AWS DocumentDB, PostgreSQL / Supabase, D1, Redis job queues, pandas, NumPy

Delivery — Requirements gathering, solution scoping, proof-of-concept builds, milestone planning, stakeholder communication, technical documentation

Front-end & Immersive — React, Next.js, krpano, WebGL, Three.js, WebXR, PTGui, Blender

EXPERIENCE

IGOTOPIA (Gecko Digital) — Malaysia

Mar 2022 – present

Virtual-tour SaaS for luxury hospitality on AWS Singapore: Node/Express API, React TypeScript CMS, Next.js + WebGL 360° viewer.

Platform & AI Engineer, Product Owner · Dec 2023 – present

Remote from Vietnam · sole engineer-owner of the Lushlist platform since Apr 2026

- **Own the platform end to end** — production and staging, four repositories, 158 commits in five months. Also act as product owner: I decompose the work and direct two junior developers and an operations colleague.
- **Built a function-calling agent that runs Python against the production CMS API.** Non-engineers now do in conversation what used to need a developer.
- **Deployed a self-hosted RAG support agent** — AnythingLLM on Docker, retrieval over the company knowledge base, documents and model serving inside our own infrastructure.
- **Scripted six luxury properties into production** — Round Hill, Sri Panwa, Shangri-La Le Touessrok, Patina Maldives, Nanuku Fiji, Six Senses Fiji. Built a per-tour customization service on Cloudflare Pages so client-specific behaviour ships without forking the core product.
- **Ran discovery, scoping and delivery for a white-labelled build** with a 3–5 person external team at Panoee. Gathered the requirement, wrote the specification and milestone plan, ran the design reviews, accepted delivery on the client side.
- **Replaced edit-on-production releases with reversible ones** — built the first staging environment (isolated EC2, database, S3 prefixes, Redis) and GitHub Actions CI/CD across three applications with health-gated deploy. Wrote ~35 runbooks so the platform stopped depending on one person.
- **Built zero-cost production monitoring with guarded self-healing** — 60-second on-box sentinel, dead-man's switch, Telegram alerting. Ran it alert-only for a week, then enabled automatic recovery behind a five-layer guard stack.
- **Led a 34-finding security audit of the production API.** Every finding carried file:line evidence, a fix and a blast-radius rating. I drove the top 11 (4 critical, 6 high) into reversible fixes and closed a credential exposure spanning seven classes of secret.

krPano Developer & Virtual Tour Analytics · Mar 2022 – Dec 2023

On site in Bali, Indonesia — the group's Indonesian branch (PT Panorama Nusantara)

- Built Python automation and data pipelines for panorama processing and publishing.

- **Cut virtual-tour load time to ~3.1s.** Tours were loading anywhere between ~4.7s and 7s with no CDN in place. Benchmarked hosting infrastructure and player settings against the PageSpeed Insights API, analysed the runs in pandas / NumPy / SciPy to isolate what moved Largest Contentful Paint, and tuned delivery to ~3.1s against a 2.5s target — acceptable for heavy interactive 3D content.
- Instrumented krpano tours with GA4 / GTAG event tracking and reported visitor engagement to client accounts.
- Developed custom krpano features and JavaScript plugins to client requirement.

WhatsApp Sales Agent — live campaign

May 2026 · Contract work for an online marketing-course business

Ran a live outbound campaign end to end: prompt and knowledge base, structured intent classification, the amoCRM / Wazzup lead pipeline, paced sending, fault diagnosis and all measurement. It ran on the client's third-party orchestrator, not on my own code.

- **Measured what it did** — 13 active days, 3,271 messages, 471 leads, 223 drawn into conversation. **99.5% of lead turns answered, 98.3% resolved with no human**, one escalation per 57 turns.
- **Published the corrected number, not the flattering one.** I separated "a reply went out" from "no human was needed", then found the escalation keyword inflated the count ~20% — the agent opened its own clarifying questions with the same phrase. I filtered it and republished.
- **Turned escalations into a closing backlog** — every question reaching a human logged as a knowledge gap with an open/resolved state; ~17–19 topics, each closing once answered.
- **Made the agent declare its own blind spots** — structured JSON carrying buy/stop intent, a confidence grade and a knowledge-gap flag, so unanswerable questions surface as data.

marketing-worker — owned rebuild

Aug 2026 – present · Independent project

Rebuilt the orchestrator as code I own: Cloudflare Workers and D1, lead state machine, versioned prompts, paced sending and nudge scheduling, and an LLM provider ladder — Claude primary, OpenAI and Kimi as failover, provider and model stamped on each message. 75/75 tests, typecheck clean; not yet carrying live traffic.

Polyglot Live — polyglot-talk.ai.studio

2026 · Built for **AI Riser Vietnam 2026** — real-time speech translation for live talks: auto-detected speech in, translated audio and text out, shared to attendees' phones or a venue screen from one session link. Built on Gemini.

360° Pipeline Developer — 3DFAB Scientific & Technical Center

Dec 2019 – Jan 2022 · Saint Petersburg, Russia

360° production for **Russian Railways' staff training centres**: interactive virtual tours of locomotives and rolling stock, ~100 panoramas per unit, openable doors and panels revealing engine components, one to two units a month on my side.

- **Owned post-production at that volume** — RAW conversion, light and colour correction, HDR merging, stitching to client-ready output.
- **Built a Python/OpenCV pre-stitching tool.** It matched control points, rejected matches outside a difference threshold, and emitted a prepared PTGui project for an operator to finish — turning the manual step from full stitching into a review pass.
- Built the krpano tours delivered to the training centres, including the state switching behind the openable panels.

Reference available from Konstantin Konstantinov, founder and CEO, 3DFAB.

Freelance — 360° Photographer & krPano Developer

Sep 2015 – Dec 2019 · Saint Petersburg, Russia

- Delivered panoramic capture, stitching and tour development direct to end clients, owning each engagement from brief to published tour.
- Built krpano tours with custom HTML/CSS/JavaScript interfaces and instrumented them for analytics, including a two-year engagement with IT Piter (2017–2019).

EDUCATION & CERTIFICATION

Yandex Practicum — Data Scientist programme · Mar 2022 – Dec 2023

Machine learning, statistics, data analysis and visualisation in Python.

Bonch-Bruевич Saint Petersburg State University of Telecommunications · 2000 – 2004

Diploma with Honours — Technician, Multichannel Telecommunication Systems.

Google Analytics Certification · 2024

LANGUAGES

Russian — native · English — professional working proficiency

INTERESTS

Data visualization · Photogrammetry · Embedded electronics (Arduino, ESP32, Raspberry Pi Pico) · 360° and landscape photography · Hiking and long-distance trekking · Vipassana meditation